

Marine DELVALLEZ

PhD student at IRISA, my research focuses on Artificial Intelligence in Education. Curious, motivated, and conscientious, I am passionate about knowledge and value sharing it.

PERSONAL INFORMATION

Date, place of birth	16 April 2004 (21 years old), Sens, France
Nationality	French
Address	9 Boulevard d'Armor, 22300 Lannion, France
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E–Mail	`madelvallez.pro@gmail.com`, `marine.delvallez@irisa.fr`
Transportation	Category B driving licence and car

EDUCATION

PhD in Computer Science October 2026 – present

Institut de Recherche en Informatique et Systèmes Aléatoires (IRISA)

Enhancing AI-Generated Adaptive Feedback with Pedagogical Strategies Informed by Predictions in Computer Science Education (Amélioration du feedback adaptatif généré par IA par intégration de stratégies pédagogiques informées par des prédictions dans le cadre de l'apprentissage de la programmation informatique)

Director : Zoltan Miklos

Supervisors : Matthieu Branthôme, Sébastien Lallé

Secondary-School Teaching Qualification (CAPES) – Computer Science (NSI) Awarded June 2026

Passed, ranked 3rd

University Diploma – Minerve Graduate Program of Excellence September 2024 – August 2026

University of Orléans

`https://www.univ-orleans.fr/fr/minerve/`

Master's Degree in Distributed Applications, Artificial Intelligence and Security (ARIAS) – Minerve track (with highest honours) September 2024 – August 2026

University of Orléans

Opening modules followed : Propriété Intellectuelle et Sciences Ouvertes (Unité Minerve Spécifique), Défi Appel à Proposition de Projet (Unité Minerve Spécifique), Fouilles de données 1 et 2 (Master de Mathématiques), Apprentissage pour l'image (Master de Mathématiques), Expériences numériques (Master de Physique), Data Science pour la physique (Master de Physique), Image Formation and Processing (Spring School Minerve), Chemo-informatique (Graduate School Orléans Numérique - GSON), Nouvelles Technologies sous R (Graduate School Orléans Numérique - GSON), Ethnographie de la communication (Master de Sciences du Langage), Dynamique de groupes (Master de Psychologie du sport), Natural Resources Management and Geopolitics (Double Licence Économie Gestion)

Bachelor's Degree in Computer Science (with highest honours) January 2023 – July 2024

University of Orléans

Computer Engineering track

Preparatory Classes for the Grandes Écoles September 2021 – December 2022
Lycée Descartes (Tours)

- 2nd year: Mathematics, Physics, Computer Science (with honours)
- 1st year: Mathematics, Physics, Engineering, Computer Science, German as a second foreign language (with honours)

French General Baccalauréat (with highest honours) July 2021
Lycée Saint Paul Bourdon Blanc

- Specialisations: Mathematics, Digital Sciences and Computer Science (NSI), Engineering Sciences (SI)
- European section

RESEARCH EXPERIENCE

Interprétabilité mécanistique pour les systèmes de *Retrieval Augmented Generation (Mechanistic Interpretability for Retrieval-Augmented Generation Systems)* February – August 2026
Orléans Laboratoire d'Informatique fondamentale d'Orléans (LIFO) – Constraints and Learning Team (CA)

- Supervisor: Anaïs Halftermeyer
- Six-month research internship (Semester 10)

Development of the *ARTESU – an ARtificial inTElligence chatbot for guiding Students facing difficulties during their University journey* project September 2025 – August 2026
University of Orléans & LIFO

Following the *Project Proposal Challenge* module, A colleague and I, sought to bring the ARTESU project to fruition. It aims to create a generative-AI-based chatbot that can guide students towards available support services according to the financial, social, mental-health, or medical difficulties they disclose.

État de l'art de l'Éxplicabilité de l'Intelligence Artificielle et ses applications au Traitement Automatique de la Langue Naturelle (State of the Art in Explainable Artificial Intelligence and Its Applications to Natural Language Processing) September 2025 – January 2026
LIFO – CA Team

- Supervisor: Anaïs Halftermeyer
- Four-month research immersion project (Semester 9): literature review and paper presentation

Synchronisation de la fonction de complexité en facteur des mots infinis dans les systèmes de numération (Synchronisation of the Factor Complexity Function of Infinite Words in Numeration Systems) March – June 2025
LIFO – Graphs, Algorithms and Models of Computation Team (GAMoC)

- Supervisor: Nicolas Ollinger
- Four-month research internship (Semester 8): literature review, proofs, and work with Walnut

État de l'art de l'Éxplicabilité de l'Intelligence Artificielle appliquée à la Macro-Économie (State of the Art in Explainable Artificial Intelligence Applied to Macroeconomics) October 2024 – February 2025
LIFO – CA Team & Orléans Economics Laboratory (LEO)

- Supervisors: Anaïs Halftermeyer & Matthieu Picault
- Collaborative project (M-DiLL) (Semesters 7 and 8): literature review, poster presentation, and a three-minute pitch

Exploration de la Logique Linéaire, de ses représentations et de ses applications (Exploring Linear Logic, Its Representations and Applications) April – July 2024
LIFO – Languages, Models and Verification Team

- Supervisor: Jules Chouquet
- Four-month undergraduate research internship (Semester 6): literature review and participation in the Linear Logic and Lambda Calculus working group

TEACHING AND SCIENCE OUTREACH

Facilitation of Introductory Computer Science Events

- Girls Can Code, 2023 and 2025 editions in Orléans
- Fête de la Science 2025

Academic Support for First-Year Students (40 hours) September – December 2024
University of Orléans
Supporting learning in first-year undergraduate computer science modules

Homework Tutoring September 2023 – July 2024
Home-based homework tutoring for lower-secondary school students

LANGUAGES

French (native)
English (≈ B2)
German (≈ A2)

TECHNICAL AND SCIENTIFIC SKILLS

Programming Languages	Python, Java, R, C/C++, OCaml, Haskell, Walnut, Rocq, Bash
Authoring Tools	Office suite and equivalents, LaTeX, web development tools (HTML, CSS, JS, PHP), Markdown and related tools (R Markdown, Quarto, Marp)

OTHER ACTIVITIES

Volunteering and Student Life	Founder and President of the Minerve Student Association of Orléans (AEMO) Help at running the restaurant at the annual Donnery festival
Interests	Contemporary dance, modern jazz dance, reading, walking, ...

PUBLICATIONS AND PRESENTATIONS

Presentation at a Team Seminar July 2026

LIFO – Constraints and Learning Team

Presentation of ongoing internship work and discussion of potential improvements and future directions.

Paper Presentation November 2025

LIFO – Constraints and Learning Team

Critical presentation of *Chefer, H., Gur, S., & Wolf, L. (2021). Transformer Interpretability Beyond Attention Visualization (No. arXiv:2012.09838). arXiv.* https://doi.org/10.48550/arXiv.2012.09838

Paper Presentation July 2024

LIFO – Languages, Models and Verification Team

Presentation of *Baillot, P. (2015). On the expressivity of elementary linear logic: Characterizing Ptime and an exponential time hierarchy. Information and Computation, 241, 3–31.* https://doi.org/10.1016/j.ic.2014.10.005

CAREER GOALS IN RESEARCH

I am particularly interested in pursuing a career as a university lecturer and researcher. It is important to me not only to create and discover new knowledge, but also to make it accessible and share it with others. Alongside my PhD, I am completing the internship associated with my success in the CAPES NSI examination. In the longer term, I am considering taking the competitive examination for the Agrégation in Computer Science.

I am currently working on Artificial Intelligence in Education and have a broader interest in human-centred and trustworthy AI, particularly explainability and interpretability in AI and Natural Language Processing. I am also interested in highly theoretical and formal topics, which I encountered during my earlier research internships.